Spacer composition, not motif grammar, drives regulatory signals in DNA sequence models

Background

Whether enhancer activity is governed by motif arrangement or by more permissive logic remains a central question in cis-regulatory biology [1,2]. We introduce the **billboard model**, a formal framework positing that regulatory activity depends on which transcription factor binding sites are present—their absolute identity and composition—not on their relative positions, orientations, or spacing. Under the billboard model, what DNA foundation models have actually learned is absolute context (compositional features) rather than positional context (relational arrangement between regulatory elements). This is a deliberately contrarian position: the prevailing view assumes that motif grammar carries regulatory meaning and that foundation models learn positional syntax. We derive testable predictions from the billboard model and evaluate them using a spacer-factored grammar sensitivity framework (SF-GSI) that isolates positional context from absolute context—resolving a critical confound in standard vocabulary-preserving shuffles, which perturb both simultaneously.

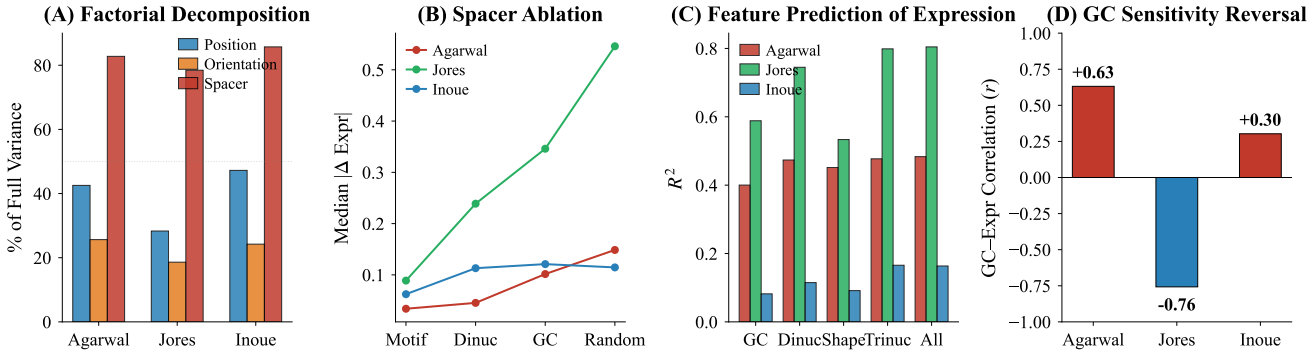

Figure 1: Absolute context dominates apparent grammar effects. (A) Factorial decomposition of prediction variance into position, orientation, and spacer components across three MPRA datasets (DNABERT-2); spacer composition consistently accounts for the largest share, supporting the billboard model. **(B)** Spacer ablation series: replacing spacer DNA (dinucleotide shuffle, GC shift, random replacement) produces larger expression changes than motif-only rearrangement, demonstrating that absolute context—not positional context—drives model sensitivity. **(C)** Feature-based regression (R^2) predicting model expression output from compositional sequence features (GC content, dinucleotide frequencies, DNA shape, trinucleotide counts); compositional features alone explain 40–80% of variance. **(D)** GC–expression correlation reverses sign across datasets, illustrating that models learn dataset-specific compositional biases rather than universal positional grammar.

Methods

SF-GSI computes a grammar sensitivity index, $GSI = \sigma(\hat{y})/|\mu(\hat{y})|$, over vocabulary-preserving motif shuffles and factors the response into positional versus absolute context using matched-spacer perturbations and null-normalized testing. The key prediction derived from our billboard model is concrete: if models encode only absolute context, then controlling for spacer composition should collapse apparent grammar sensitivity to noise—i.e., the residual positional signal should be statistically indistinguishable from a null distribution.

The framework proceeds in three stages. First, a *factorial decomposition* isolates the contributions of motif position, motif orientation, and spacer composition by generating matched perturbation sets in which each factor is varied independently while the others are held fixed; effect sizes are expressed as the fraction of full-shuffle variance attributable to each factor. Second, a *spacer ablation series* applies graded compositional perturbations—motif-only rearrangement, dinucleotide-preserving shuffle, GC-content shift, and full random replacement—to quantify the dose–response relationship between compositional disruption and model prediction change. Third, a *feature decomposition* regresses model predictions against explicit compositional features (GC content, dinucleotide frequencies, trinucleotide counts, and DNA shape parameters) to estimate how much of the expression signal is explained by absolute context alone. Statistical significance is assessed via a three-stage correction cascade: an initial F-test on raw GSI, followed by z-score null normalization against matched shuffles, and finally Benjamini–Hochberg FDR correction at $\alpha = 0.05$. Positive controls use synthetic constructs with fixed spacer sequence and known orientation-dependent activity, verifying that the framework retains power to detect genuine positional effects when they exist.

We applied SF-GSI to five MPRA datasets (Agarwal, Klein, Inoue, Vaishnav, and Jores), totaling 7,650 model–enhancer measurements: 3 foundation models × 5 datasets × 500 enhancers, plus Enformer on 3 human datasets

× 50 enhancers. The model set spans transformer, state-space, and MPRA-native architectures—DNABERT-2, Nucleotide Transformer v2, HyenaDNA, Enformer, and PARM [4–7]—ensuring that conclusions are not artifacts of a single model family. Code: `https://github.com/bryanc5864/GRAMLANG`.

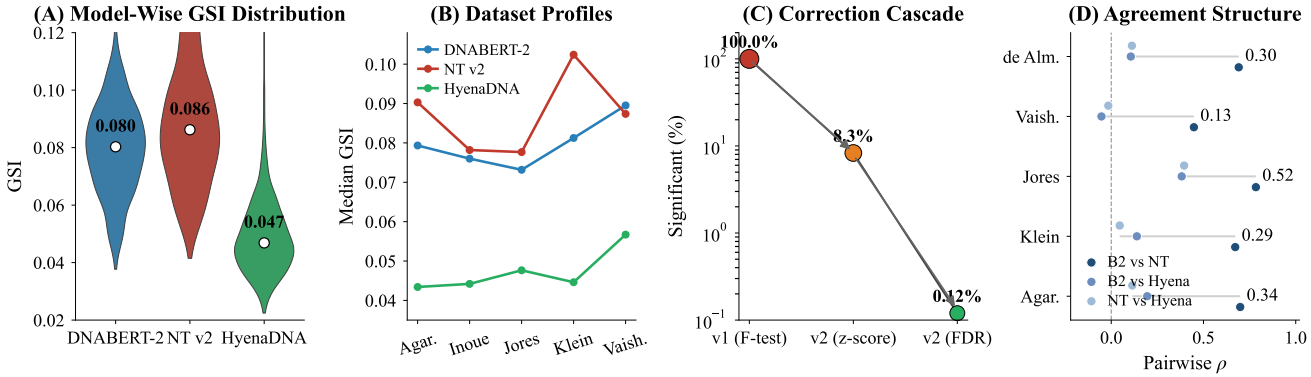

Figure 2: Statistical correction collapses apparent grammar prevalence. (A) Violin plots of raw GSI values by model (DNABERT-2, Nucleotide Transformer v2, HyenaDNA), with median points; all models show similar distributions, indicating that grammar sensitivity estimates are not architecture-specific. **(B)** Median GSI profiles across five datasets for each model; dataset identity drives more variation than model choice, consistent with the billboard view that compositional properties of each library dominate. **(C)** Correction cascade: grammar-sensitive calls fall from 100% under naive thresholding to 8.3% after null normalization and to 0.12% (9/7,650) after FDR correction—a three-order-of-magnitude collapse that matches the billboard prediction. **(D)** Pairwise cross-model GSI correlations by dataset; strong agreement in three datasets but near-zero or negative correlations in Vaishnav and Inoue, showing that residual positional signal is context-dependent rather than reflecting a universal grammar learned across architectures.

Results

Under naive thresholding all tested enhancers appear grammar-sensitive, but this falls to 8.3% after null normalization and to 0.12% (9/7,650) after FDR correction (Fig. 2). That three-order-of-magnitude collapse is precisely the signature predicted by our billboard model: once absolute context is controlled, the apparent prevalence of positional syntax vanishes. Absolute context (spacer composition) accounts for 78–86% of grammar-sensitivity variance (Fig. 1), while positional context—motif-only rearrangements—produces substantially smaller perturbation effects than spacer replacement or GC-shifting perturbations. These models have learned *what* is present, not *where* it is; the signal widely attributed to grammar is, in the vast majority of cases, a compositional signal misread as a positional one.

Positive controls confirm the framework does not erase genuine grammar: with spacer sequence held fixed in synthetic constructs, orientation effects remain strongly detectable across models ($p \leq 7.2 \times 10^{-15}$, with two models reaching numerical underflow). However, these controlled cases are rare exceptions, not the rule. Variance decompositions assign only 0–1.6% of explained variance to arrangement versus 8–22% to motif vocabulary—consistent with our billboard model's central claim that absolute identity, not positional syntax, is the dominant learned feature. Cross-species transfer is near zero, further suggesting that whatever positional logic models encode is not generalizable. Cross-model agreement concentrates in specific biological contexts rather than reflecting architecture-general positional learning, indicating that dataset provenance matters more than model design.

Conclusions

These results validate the billboard model we propose: DNA foundation models encode absolute context—what binding sites and sequence features are present—but show negligible encoding of positional context once composition is controlled. The billboard model reframes how the field should evaluate regulatory foundation models: benchmark perturbations that fail to control for absolute context systematically overstate positional knowledge. Claims of learned regulatory grammar should be made only under perturbation designs that explicitly factor out absolute context; without that control, compositional signals masquerade as positional ones.

References: [1] Arnosti & Kulkarni, *J. Cell. Biochem.* 94:890, 2005. [2] Long et al., *Cell* 167:1170, 2016. [3] Castro-Mondragon et al., *Nucleic Acids Res.* 52:D142, 2024. [4] Zhou et al., *bioRxiv* 2023, DNABERT-2. [5] Nguyen et al., *bioRxiv* 2023, HyenaDNA. [6] Avsec et al., *Nature Methods* 18:1196, 2021, Enformer. [7] Agarwal et al., *Science* 380:eadh0559, 2023; Jores et al., *Nat. Plants* 7:842, 2021.